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10.7.2 Written down value / Diminishing balance method

Under this method, depreciation is charged at a fixed percentage on the written down value of the asset every year. Hence, it is called written down value method. Written down value is the book value of the asset, i.e., original cost of the asset minus depreciation upto the previous accounting period. As the amount of depreciation goes on decreasing year after year, it is called diminishing balance method or reducing installment method.

The following formula is used to compute the rate of depreciation under written down value method:



$$\left[1 - \sqrt[n]{\frac{\text{Scrap value}}{\text{Original cost}}}\right] \times 100$$

Example: Original cost ` 1,00,000; Scrap value ` 1,000; Useful life 5 years

$$\left[1 - \sqrt[5]{\frac{1{,}000}{1{,}00{,}000}}\right] \times 100 = [1-(0.01)^{1/5}] \times 100$$

$$= (1 - 0.3981) \times 100$$

$$= 0.6019 \times 100 = 60.19\%$$

Rate of depreciation = 60.19%

If the scrap value is 10,000; rate of depreciation is 36.90%

If there is no scrap value, rate of depreciation will be 100%. Hence, to calculate depreciation the scrap value is taken as 1.

If the scrap value is less, rate of depreciation will be high.

Example

On 1.1.2012, a firm purchased a machine at a cost of ` 1,00,000. Depreciation charged at 10% p.a. on written down value method for the five years is as follows:

Cost of the asset on 1.1.2012	` 1,00,000
Less: Depreciation for 2012 at 10% on 1,00,000	` 10,000
Written down value on 1.1.2013	` 90,000
Less: Depreciation for 2013 at 10% on 90,000	` 9,000
Written down value on 1.1.2014	` 81,000
Less: Depreciation for 2014 at 10% on 81,000	` 8,100
Written down value on 1.1.2015	` 72,900
Less: Depreciation for 2015 at 10% on 72,900	` 7,290